

The rules of differentiation

Lessons 7–9 · Algebra 11, §1.4 · P1 · 7.2–7.4 · 28 problems

Differentiate with respect to x . No chain rule is needed anywhere on this sheet.

NAME

CLASS

DATE

Easy 7

1

$(x^9)'$

2

$(7x^3)'$

3

$(x^5 - x^2)'$

4

$(-12)'$

5

$(6x + 1)'$

6

$(\frac{1}{x^3})'$

7

$(2\sqrt{x})'$

Medium 7

8

$(5x^3 + 2x^2 - 8x + 1)'$

9

$(\frac{4}{x^5})'$

10

$(x^2\sqrt{x})'$

11

$((x - 2)(x + 6))'$

12

$(3x(x^2 - 4))'$

13

$(\frac{x + 3}{x - 1})'$

14

$(\frac{x^3 - 5x}{x})'$

Hard 7

15

$((3x + 2)(x^2 - x))'$

16

$(\frac{x^2 + 1}{x - 3})'$

17

$(x^4\sqrt{x})'$

18

$(\frac{x + 4}{\sqrt{x}})'$

19

$(x^3(x - 2)(x + 2))'$

20

Find y'' for $y = 2x^5 - x^3 + 4x$

21

Find the values of x at which $y = x^3 - 12x$ has gradient 15

Very hard 7

22

$((2x - 1)(x + 1)(x + 3))'$

23

$(\frac{(x - 1)(x^2 + 4)}{x^2})'$

24 Find the gradient of $y = \frac{2x + 1}{x^2 + 1}$ at $x = 0$

25 Find the fourth derivative of $y = \frac{1}{x^2}$

26 Differentiate $y = (x^2 - 3)^2$ **without** using the chain rule

27 $y = ax^2 + bx$ passes through $(2, 2)$ and has gradient 5 at $x = 2$. Find a and b .

28 The tangent to $y = x^2 + \frac{k}{x}$ at $x = 1$ is horizontal. Find k , and the value of y there.
